RiskLens

AI-Powered Credit Risk Intelligence Platform

Developed by: Gourinandini

Streamlit · LightGBM · SHAP · Groq LLaMA 3.3

Use Case & Business Value

The Credit Risk Challenge

• High Default Exposure:
Financial institutions face heavy losses from unpaid loans.

• Opposing Objectives:
Underwriters need accurate predictions, but legacy models lack granular resolution.

• Black-Box Compliancy Issues:
Modern ML (e.g., LightGBM) yields high performance but remains unexplainable, failing regulatory audits (like SR 11-7).

• Data Inaccessibility:
Risk managers struggle to explore portfolio metrics without writing complex SQL queries.

The RiskLens Core Solution

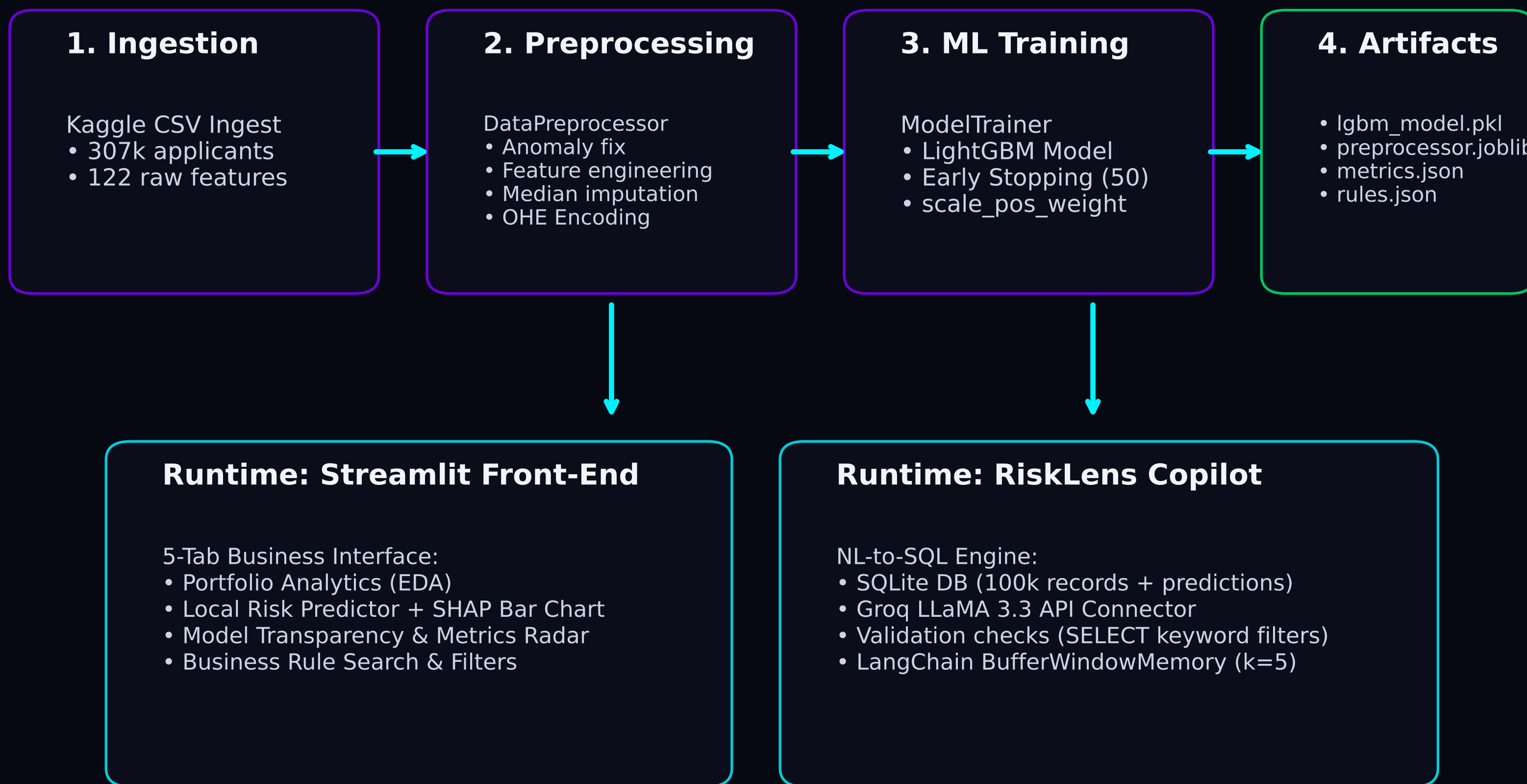
✔ Calibrated Risk Predictor:
Real-time default probability scoring using optimized thresholds.

✔ Regulatory Auditability via Rules:
Post-hoc surrogate decision tree maps complex predictions into human-readable if/then decision rules.

✔ Explainability at Scale:
Global & local feature attributions using SHAP explainers.

✔ RiskLens Copilot (NL-to-SQL):
Allows business teams to query SQLite portfolio records in natural language without technical knowledge.

Platform System Architecture



Model Performance & Imbalance Strategy

Model Metrics Report

- ROC-AUC Score: 0.7683
- PR-AUC Score: 0.2591 (3.2x over random baseline)
- F1 Score: 0.3202
- Calibrated Threshold: 0.6815
- Overall Validation Accuracy: 86.0%

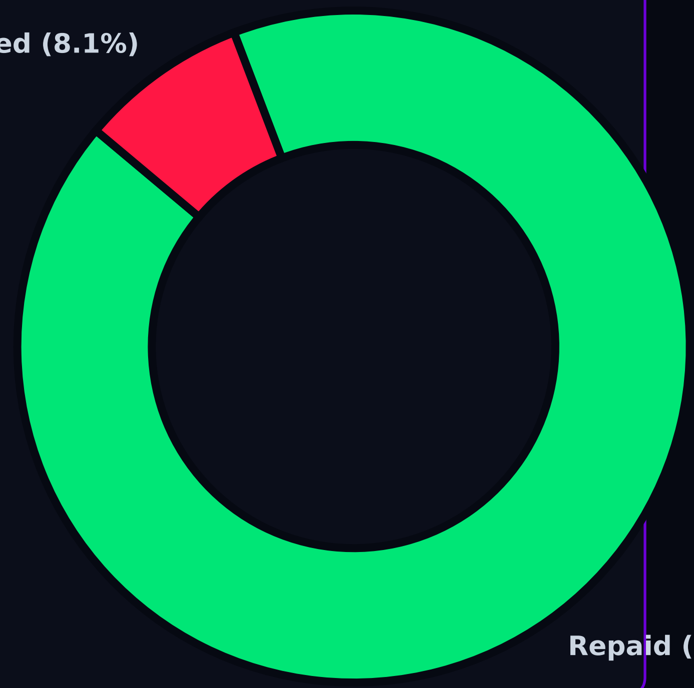
Class Imbalance Strategy

- The Challenge:
  Severe skewness (~92% repaid, ~8% defaulted). A naive model achieves 92% accuracy by approving everyone.

- scale_pos_weight Penalty:
  Loss function penalizes missed defaults ~11.3x more heavily.

- F1-Optimal Thresholding:
  Precision-recall curve search anchors high-risk classification at score >= 0.6815 to maximize minority-class F1.

Applicant Default Imbalance (307k)



Model Transparency & Rules Engine

SHAP Local & Global Explanations

• True Transparency:
RiskLens uses shap.TreeExplainer to compute exact Shapley attribution values for applicant features.

• Local Risk Shift:
Attributions show how each feature pushes individual probability higher (red, increases risk) or lower (green, reduces risk) relative to the dataset expected base probability.

• Top Global Features:
1. External Credit Score 2 (EXT_SOURCE_2)
2. External Credit Score 3 (EXT_SOURCE_3)
3. Avg External Score (EXT_SOURCE_MEAN)
4. Age (AGE_YEARS)
5. Leverage Ratio (CREDIT_INCOME_RATIO)

Surrogate Rule Derivation

• Method:
Trains a shallow DecisionTree (max depth=4) on LightGBM binarized targets to model the complex model's boundary.

• Key Metrics:
- Extracted Rules: 16 human-readable paths
- Surrogate Fidelity: 85.73% agreement rate

• Calibrated Band Rules:
Assigns leaves to Low, Medium, and High bands based on average leaf default probability (optimal thresholds).

• Sample Rule Output (Medium Risk):
IF External Credit Score 2 <= 0.4305
AND External Credit Score 3 > 0.31
AND Employment Duration > 4.5 years
--> RISK BAND: Medium (Confidence: 58.9%, Support: 4.4%)

RiskLens Copilot — Conversational Analyst

Copilot Agent Core Capabilities

• Zero-Code Data Access:
Translates natural language questions into executable SQL queries, runs them against SQLite, and summarizes output.

• Multi-Turn Memory Context:
Utilizes LangChain ConversationBufferWindowMemory (k=5) to retain query sequence for subsequent follow-up tasks.

• Schema-Anchored Safeguards:
Injects DDL and standard category values inside prompts to block column hallucination.

• Two-Layer Injection Defenses:
- Validates query structure (forces SELECT statements only).
- Blocks destructive commands (DROP, DELETE, UPDATE, ALTER, etc.).
- Caps output fetching to 100 records to prevent memory overflow.

Verified Query Patterns & Examples

✔ Query Pattern 1: Aggregates by Class
* 'What is the average income of applicants who defaulted?'
* SQL: SELECT AVG(AMT_INCOME_TOTAL) FROM applications WHERE TARGET=1

✔ Query Pattern 2: Categorical Default Rates
* 'Show default rate by education type'
* SQL: SELECT NAME_EDUCATION_TYPE, AVG(TARGET)*100 FROM applications...

✔ Query Pattern 3: Filter & Count Queries
* 'How many female applicants have more than 2 children?'
* SQL: SELECT COUNT(*) FROM applications WHERE CODE_GENDER='F'...

✔ Query Pattern 4: Top-N Leaders
* 'What are the top 10 highest credit amounts?'
* SQL: SELECT AMT_CREDIT FROM applications ORDER BY AMT_CREDIT DESC...

✔ Query Pattern 5: Multi-band comparisons
* 'Show average external credit score by risk band'
* SQL: SELECT RISK_BAND, AVG(EXT_SOURCE_MEAN) FROM applications...

Interactive Business UI Preview

Sidebar Panel

RiskLens v2.0

Enterprise Edition

ROC-AUC: 0.7683

PR-AUC: 0.2591

F1 Score: 0.3202

● ML Model: Loaded

● Database: Connected

● Copilot: Connected

Interactive Workspace Tabs

EDA Dashboard | Risk Predictor | Explainability | Decision Rules | RiskLens Copilot

Applicants	Default Rate	Avg Credit	Avg Income
100,000	8.09%	₹5.99L	₹1.68L

Portfolio Distribution by Calibrated Risk Bands